

AIMS System - Detailed Class Design Documentation

11. Design Class: Cart

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	cartItemsList	List<CartItem>	empty	List of CartItem in the cart
2	cartID	float	0.0	Unique identifier for the cart

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	calculateTotalPrice	float	Calculates and returns the total price
2	empty	void	Empties the cart
3	checkEligibilityOfRushDeliveryOfChosenItems	boolean	Checks if all items are eligible for rush delivery
4	checkAvailabilityOfChosenItems	boolean	Checks if all items have sufficient inventory
5	isEmpty	boolean	Checks if the cart is empty

Parameter:

- checkEligibilityOfRushDeliveryOfChosenItems(province: String, district: String): boolean
 - province:** The name of the province where the delivery is requested.
 - district:** The name of the district within the province.
 - Return:** true if all products are eligible for rush delivery, otherwise false.

Exception: None

Method:

How to use parameters/attributes:

- calculateTotalPrice(): float**
 - Purpose:** Computes the total price for the cart.
 - Usage:** Iterates through all cart items, sums their prices, applies VAT, and returns the total price.
- empty(): void**
 - Purpose:** Empties the cart by removing all items.
 - Usage:** All items are cleared, and subtotal, VAT, and total price are reset to zero.
- checkEligibilityOfRushDeliveryOfChosenItems(province: String, district: String): boolean**
 - Purpose:** Determines if all products in the cart are eligible for rush delivery.
 - Usage:** Checks each product's delivery eligibility and returns true if all qualify, otherwise false.
- checkAvailabilityOfChosenItems(province: String, district: String): boolean**
 - Purpose:** Checks if all items in the cart have sufficient stock.
 - Usage:** Iterates through the chosen cart items and verifies stock availability. Returns true if all items are available, otherwise false.
- isEmpty(): boolean**
 - Purpose:** Checks if the cart is empty.
 - Usage:** Returns true if there are no items in the cart, otherwise false.

State:

- If cartItemsList is empty → Empty
- If cartItemsList contains at least one item → Active
- If checkout() is called and payment is not completed → CheckedOut
- If payment is successful → Paid
- If cart is emptied (either manually or after order completion) → Empty